Islamic University of Technology

Board Bazar, Gazipur, Dhaka

SWE 4304 – Software Project Lab-I

Final Project Report

LabTrack

Lab Management System

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SPL Team - 18

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CHAPTER 1 : PROBLEM STATEMENT & POTENTIAL SOLUTION

1.1 Problem Statement :

Modern research laboratories frequently depend on manual record-keeping or scattered digital files (such as loose spreadsheets and paper logs). This approach creates several critical problems:

- Inefficiency and time loss in locating records, scheduling resources, or tracing sample history.
- Frequent human errors in data entry, inventory tracking, and experiment documentation.
- Booking conflicts for equipment and lab rooms due to lack of real-time availability checks.
- Poor accountability and traceability because changes to experiments or protocols are not versioned or logged.
- Reduced research reproducibility caused by untracked sample lineage and inconsistent standard operating procedures (SOPs).

1.2 Potential Solution :

LabTrack solves these issues with a lightweight, centralized terminal-based Research Lab Management System. All data is stored in simple

text/CSV files for portability and zero database overhead. The system introduces role-based access control to enforce security and responsibility, Git-style version control for experiments and protocols to track every change with full history and restore capability, automated sample lineage tracking for complete genealogy, real-time inventory monitoring with low-stock alerts, and conflict-free reservation systems for equipment and lab rooms. These features collectively eliminate manual errors, ensure data integrity, prevent resource conflicts, and deliver full workflow transparency in academic and small-scale lab environments.

CHAPTER 2 : IMPLEMENTED FEATURES

2.1 Secure User Management :

The system supports multiple users with individual login accounts and role-based access control. Each user is assigned a role such as researcher, lab manager, technician, or administrator, which determines the actions they are allowed to perform. This prevents unauthorized access to sensitive operations and ensures accountability. The feature enhances system security while maintaining a clear division of responsibilities.

- Username & Password Authentication :
    - Users log in using valid credentials (Name, ID, Password).
    - Passwords are securely hashed before being stored in the CSV files.

- Role-Based Access Control:
    - Researcher: Logs experiments and uses equipment.
    - Lab Manager: Approves protocols and manages resources.
    - Technician: Updates inventory and maintenance logs.
    - Lab Assistant : handles reports and samples.
    - Admin: Full system access.

2.2 Experiment & Version Control:

The system allows users to create, update, and view experiment records in an organized manner. Each modification is saved as a new version, enabling users to track changes over time. Previous versions can be reviewed or restored when needed, ensuring data integrity. This feature helps maintain accurate experimental records and supports reproducibility.

- Create and update experiment records.
- Maintain multiple versions of experiments.
- View previous versions and restore if needed.
- Track histories of who made changes.
- This feature is inspired by Git-style version control to prevent data loss.

2.3 Protocol Management :

Standard operating procedures (SOPs) can be stored, updated, and managed within the system. Lab managers are responsible for approving or modifying protocols to ensure compliance with laboratory standards. The system maintains version history for protocols, allowing users to track changes. This feature ensures consistent and approved procedures are followed during experiments. Store Standard Operating Procedures (SOPs). Managers can approve, edit, or update protocols. Version history maintained for protocol changes. This feature helps ensure standardized and approved lab practices.

2.4 Sample Lineage Tracking:

The system tracks the relationship between parent samples and derived samples throughout the experiment lifecycle. Users can trace a sample's origin, transformations, and usage across experiments. This improves traceability and research reliability. The feature is essential for maintaining accurate experimental documentation.

- Track parent and child samples.
- Maintain full genealogy of samples.
- Trace sample origin to final usage. This feature helps boost research accuracy and reproducibility.

2.5 Inventory Management :

Inventory items such as reagents and materials are monitored with updated stock levels. Technicians can log usage, updates, and maintenance activities. The system helps prevent shortages and reduces material wastage. This feature ensures efficient resource utilization within the laboratory.

- Track real-time quantity of reagents.
- Log material usage per experiment.
- Technician-managed inventory updates.

2.6 Equipment & Lab Reservation:

The system allows users to reserve laboratory equipment and rooms for specific time periods. It prevents scheduling conflicts by validating availability before confirming bookings. Usage and maintenance logs are maintained for accountability. This feature improves resource planning and operational efficiency.

- Reserve equipment for specific time slots.
- Lab room booking
- Prevent booking conflicts.
- Track equipment availability.
- Maintain maintenance logs

CHAPTER 3 : TOOLS & TECHNOLOGIES

LabTrack is built using a simple yet robust technology stack suitable for a terminal-based academic project:

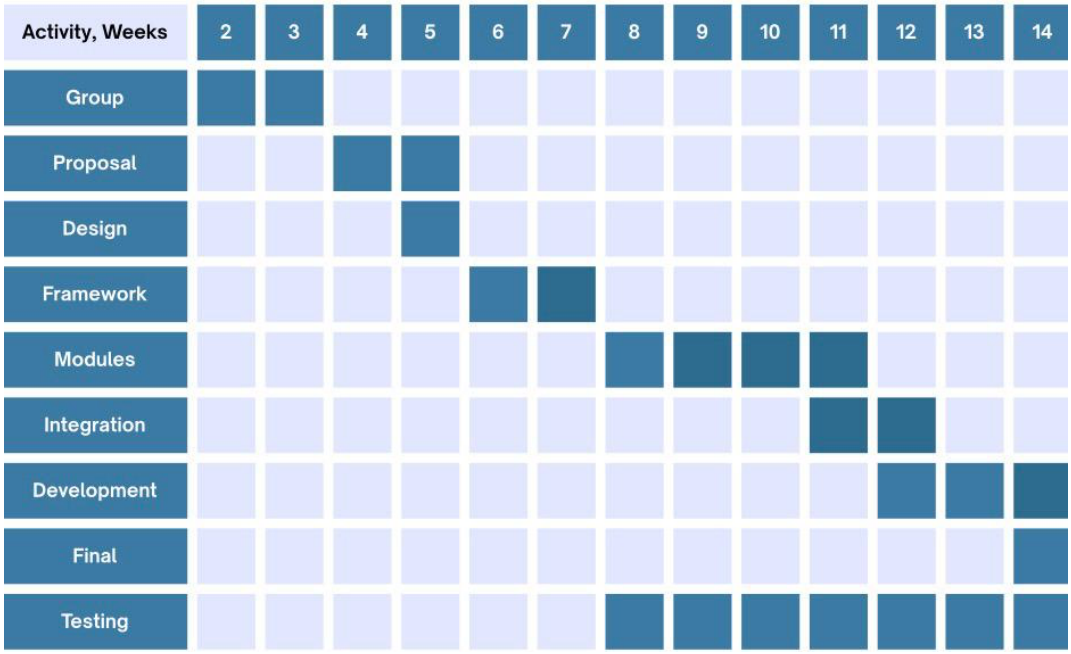
- Programming Language: Java — chosen for strong object-oriented design, excellent file-handling capabilities, and platform independence.
- Development Environment: Visual Studio Code and IntelliJ IDEA for efficient coding, debugging, and project organization.
- User Interface: Pure command-line terminal interface with text-based menus for lightweight interaction (no graphical dependencies).
- Data Storage: Structured text files (CSV and TXT formats) — ensures simplicity, easy portability, and compliance with project constraints without requiring any database.

- Operating Systems: Fully compatible with Windows, MacOS and Linux terminals.
- Testing Method: Manual testing using sample test datasets to verify login, file I/O, module interactions, and end-to-end functionality.

CHAPTER 4 : APPLICATION DOMAIN

LabTrack belongs to the domain of Research Laboratory Management Systems, specifically designed for academic institutions and small-scale research labs. It targets daily operational needs such as experiment documentation, sample management, protocol compliance, inventory control, equipment and room scheduling, and maintenance logging. The terminal-based, file-driven architecture makes it ideal for environments where graphical interfaces or heavy database servers are impractical or unavailable — providing a portable, secure, and efficient solution that enhances productivity, accountability, and research reproducibility in university research labs.

CHAPTER 5 : PROJECT TIMELINE



CHAPTER 6 : CONTRIBUTION

6.1 Abdullah Tiran (230042105) :

- Led the design and implementation of the Authentication module, establishing secure access flow and role-aware login handling.
- Developed the Booking module, including core booking operations and service-level workflow execution.

- Co-developed the Version Control module with Rehan, contributing to version tracking logic and integration with experiment records.
- Conceptualized and implemented the CSV-based storage approach, defining the project's lightweight persistent data strategy.
- Implemented core console interaction architecture, including command-driven execution flow and console-layer behavior in the main runtime path.
- Built key utility components, specifically FileManager and InputHelper, to standardize file operations and validated user input handling.
- Contributed to the Admin class alongside Rehan and Shudipto, supporting shared administrative control capabilities.
- Implemented major LabManager-side functionality for operational supervision and system-level management tasks.

6.2 Rufayed Ul Alam (230042121) :

- Designed and implemented the Experiments module, including Experiment and ExperimentService for structured experiment lifecycle handling.
- Developed Colors and TablePrinter to improve console readability, output consistency, and structured tabular presentation.

- Implemented Researcher role functionality to support researcher-specific interaction patterns and permissions.
- Developed foundational User-layer behavior to strengthen role hierarchy and shared identity abstractions.
- Co-developed the Version Control module with Tiran, contributing to core logic design and feature integration.
- Contributed to the Admin class as a shared ownership component with Tiran and Shudipto.

6.3 Shudipto Sarwar (230042142) :

- Implemented the Inventory module components, including InventoryItem, InventoryService, and ItemService, covering item modeling and operational inventory workflows.
- Developed LabRoom to support laboratory space representation and room-level operational data handling.
- Implemented Protocol and ReportService, enabling protocol management and report-oriented service processing.

- Built Sample and SampleService to support sample modeling, service logic, and sample lifecycle operations.
- Implemented remaining role-side components, including LabAssistant and Technician, completing operational role coverage.

CHAPTER 7 : FUTURE WORK PLAN

7.1 Protocol management completion :

Add full protocol workflows: create, edit, approve/reject, and protocol history tracking.

7.2 Lab room management completion :

Add room availability view, booking conflict checks, cancellation, and rescheduling.

7.3 Reporting module expansion :

Move from basic counts to role-based operational reports, usage trends, and exportable summaries.

7.4 Authentication and security hardening :

Upgrade to secure password handling, reset/change flows, failed-login limits, and activity/audit logs.

7.5 Full borrow and booking lifecycle :

Cover complete states: request, approve/reject, fulfill, return, overdue handling, close case.

7.6 Data layer evolution :

Gradually move from CSV persistence to database-backed storage for reliability and scale.

7.7 Platform Direction Options (CLI, MERN, Or More) :

- Advanced CLI product (short-term, low risk)
- Keep Java terminal app, add cleaner command flows, better validation, logs, and analytics screens.

7.8 MERN web app (modern UI path) :

Build React frontend + Node/Express API + MongoDB, while reusing existing business rules as reference.

CHAPTER 8 : SOURCE REPOSITORY

Githib Repository :
https://github.com/notexceptional/LABTRACK/tree/master?files=1